

CAREER: Foundations of Ecologically Intelligent Computing Systems

Yi Ding, yiding@purdue.edu

Overview

Artificial intelligence (AI) and high-performance computing now operate at a scale where digital decisions increasingly translate into physical consequences. Training models, operating datacenters, manufacturing chips, and refreshing hardware all require energy, water, land, materials, and industrial processes that can affect surrounding natural systems. In this project, **ecological impact** refers to these real-world consequences of computing infrastructure across its lifecycle, including resource depletion, ecosystem stress, pollution-related damage, and pressures created by rapid infrastructure expansion.

Despite this growing footprint, most computing systems are still optimized primarily for performance, cost, and energy efficiency, with little ability to measure or respond to broader ecological consequences. As AI demand accelerates, this gap creates strategic risks: inefficient infrastructure growth, hidden resource bottlenecks, rising operational constraints, and missed opportunities for smarter deployment.

This CAREER project will establish the foundations of ecologically intelligent computing systems, systems that can sense, model, and optimize their ecological impact alongside traditional computing objectives. Rather than treating ecological consequences as external afterthoughts, these systems will use ecological signals as actionable inputs for hardware design, workload scheduling, model deployment, and lifecycle planning. By creating the metrics, empirical understanding, and adaptive control methods needed for ecological intelligence, this project aims to enable a new generation of computing infrastructure that is high-performing, scalable, and aligned with long-term resource realities.

Keywords: AI infrastructure; lifecycle modeling; resource-aware optimization

Intellectual Merit

This project addresses a fundamental systems research question: how should computing systems be designed and operated when performance objectives must be balanced with lifecycle resource use and regional operating constraints? To answer this question, the project will pursue three integrated thrusts:

- **Metrics and Modeling Foundations for Ecological Impacts.** Develop principled metrics and lifecycle modeling frameworks to quantify ecological impacts of computing systems, capturing both embodied and operational effects across diverse deployment contexts.
- **Characterization of Modern AI Infrastructure.** Build tools and platforms to provide the first systematic understanding of ecological impacts across emerging computing workloads and platforms.
- **Adaptive Ecological Optimization.** Design schedulers and cross-layer optimization methods that dynamically improve system performance and efficiency under resource and ecological constraints.

The project advances computer systems research by introducing new objectives, benchmarks, and control mechanisms for next-generation AI infrastructure.

Broader Impacts

This project will help enable continued growth of U.S. computing capacity under emerging infrastructure limits. Expected outcomes include open-source tools, datasets, and benchmarks for academia and industry; improved methods for efficient AI and datacenter operation; and decision support for future infrastructure planning. Educational activities will integrate research into new coursework on AI systems and real-world infrastructure constraints, train graduate and undergraduate students in cross-disciplinary systems research, and broaden participation through research experiences and outreach. By preparing the next generation of engineers to design computing systems that align with physical resource realities, the project supports long-term national competitiveness and technological leadership.